BVAR vs. Information Criteria: A Monte Carlo Simulation Study

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June 16, 2026

Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Keyword 1, Keyword 2, Keyword 3
JEL Codes: C15, C32

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1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

2 Methodology

2.1 Data Generating Process

Explain the true DGP based on Kilian and Murphy (2014) or other literature.

2.2 Optimal Prior Selection via Marginal Data Density

Describe how the shrinkage parameter (τ) is treated as an optimization problem rather than a fixed heuristic.

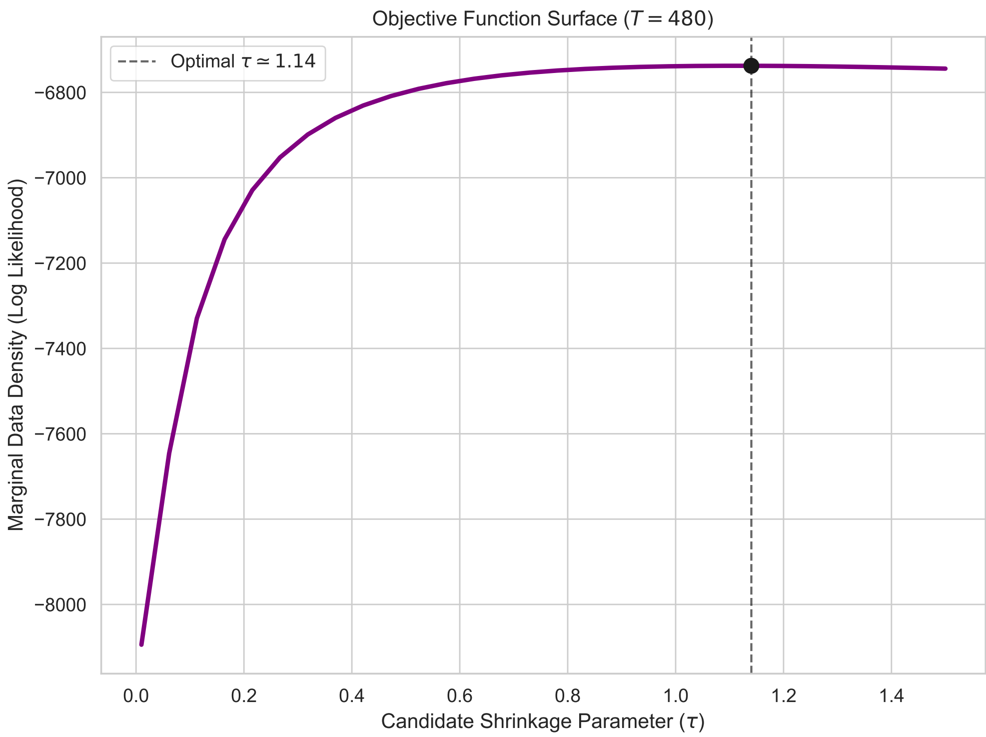

Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3 Simulation Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

3.1 Behavior of Standard Information Criteria

Before evaluating the Bayesian estimators, we examine the baseline lag detection performance of the classical Information Criteria.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by standard Information Criteria.

p_0	Estimator T	AIC	AICc	SIC (BIC)	HQC
4	96	9.85	1.00	1.29	1.92
	144	2.65	1.00	1.51	2.02
	240	3.12	1.00	1.98	2.05
	480	3.51	1.00	2.01	2.34
6	96	10.81	1.00	1.22	2.02
	144	3.49	1.00	1.53	2.05
	240	3.97	1.00	1.98	2.04
	480	5.69	1.00	2.00	2.63
8	96	11.21	1.00	1.26	2.26
	144	5.04	1.00	1.51	1.96
	240	6.01	1.00	1.98	2.04
	480	8.01	1.00	2.00	2.68
10	96	11.47	1.00	1.20	2.66
	144	4.86	1.00	1.55	2.03
	240	6.72	1.00	1.96	2.08
	480	8.79	1.00	2.01	2.59

3.2 The Shrinkage Strategy: Priors vs. Data

This section compares hard lag selection against dynamic BVARs. We analyze both the bias-variance tradeoff and how the optimal prior relaxes asymptotically as sample sizes increase.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage.

T	p_0	AIC	AICc	SIC (BIC)	HQC	BVAR-RW ($\tau = 0.05$)	BVAR-RW ($\tau = 0.20$)	BVAR-RW ($\tau = 0.50$)	BVAR-WN ($\tau = 0.20$)	SOTA-BVAR (WN) (Opt τ)	SOTA-BMA (WN) (Opt τ)
96	4	1.328	0.957	0.985	0.998	190.036	4.063	1.580	1.817	**0.929**	0.933
	6	1.426	0.913	0.886	1.028	210.207	4.952	1.736	1.984	0.920	1.001
	8	1.267	0.768	0.807	0.875	184.654	3.934	1.550	1.665	0.980	0.857
	10	1.283	0.855	0.829	0.800	158.277	3.384	1.416	1.447	0.787	**0.782**
144	4	1.106	1.206	1.145	**1.085**	88.872	2.753	1.376	1.655	1.335	1.245
	6	1.014	1.017	1.086	1.050	67.452	1.770	1.079	1.116	1.120	**0.984**
	8	0.908	0.918	0.920	**0.869**	60.869	1.745	1.048	1.120	1.016	1.011
	10	0.805	0.826	0.730	**0.719**	56.227	1.712	0.951	0.994	0.809	0.847
240	4	1.052	1.042	1.067	1.056	25.208	1.159	1.017	1.074	**0.890**	1.020
	6	1.211	1.018	1.026	**1.000**	30.403	1.510	1.116	1.251	1.419	1.321
	8	0.986	1.030	0.943	0.968	20.176	1.047	0.931	1.003	0.942	**0.915**
	10	**1.062**	1.236	1.289	1.249	29.587	1.909	1.275	1.408	1.481	1.312
480	4	0.956	1.589	1.214	1.091	9.532	1.150	0.946	1.074	1.063	1.248
	6	**0.905**	1.148	1.210	1.107	9.284	1.447	1.048	1.160	1.312	1.262
	8	0.999	1.218	1.307	1.122	8.244	1.011	**0.925**	0.934	1.126	0.980
	10	0.939	1.008	1.157	1.036	8.927	1.378	1.187	1.267	1.080	1.210

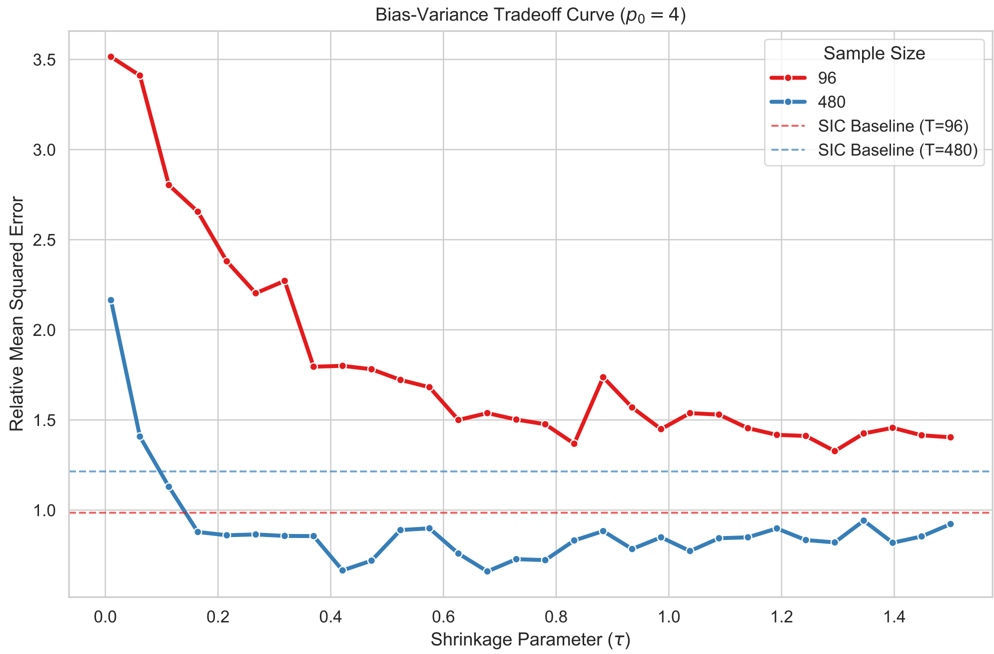

Figure 2: Bias-Variance Tradeoff Curve for SVAR Estimation. Demonstrates the out-of-sample forecasting error across varying degrees of shrinkage (τ) for small and large sample sizes.

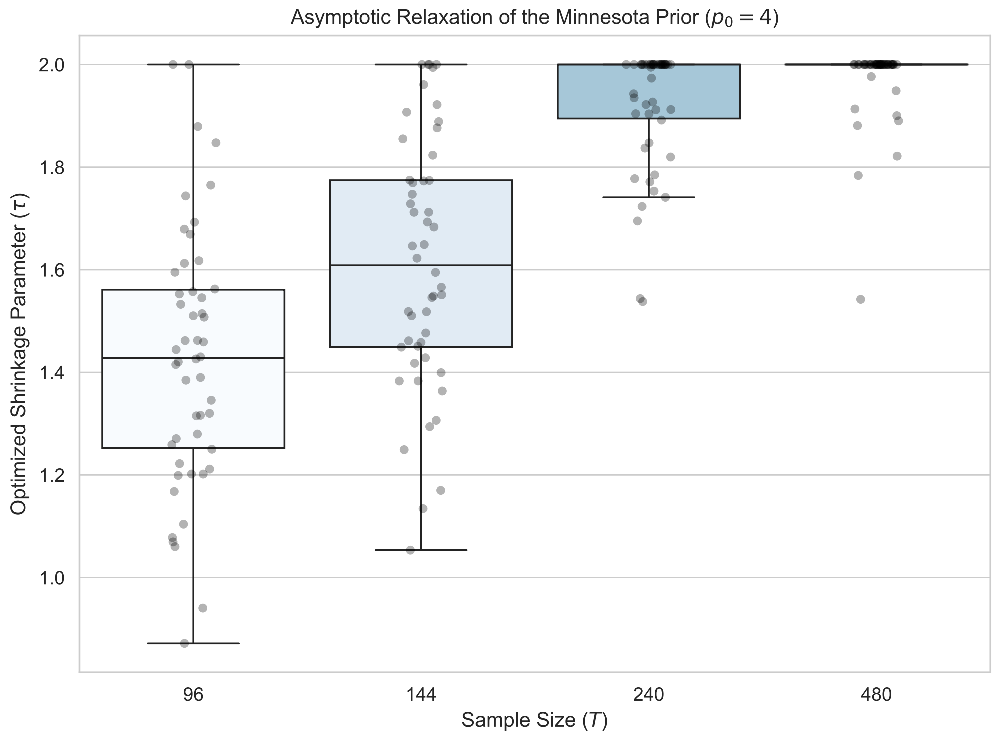

Figure 3: Asymptotic Relaxation of the Minnesota Prior. As the sample size (T) increases, the optimizer naturally selects looser priors, letting the data dominate the estimation.

3.3 The Uncertainty Strategy: Averaging vs. Selection

Here, we evaluate True Bayesian Model Averaging (BMA) against Information Criteria. We focus on how BMA acts as an insurance policy against catastrophic lag misspecification in small samples.

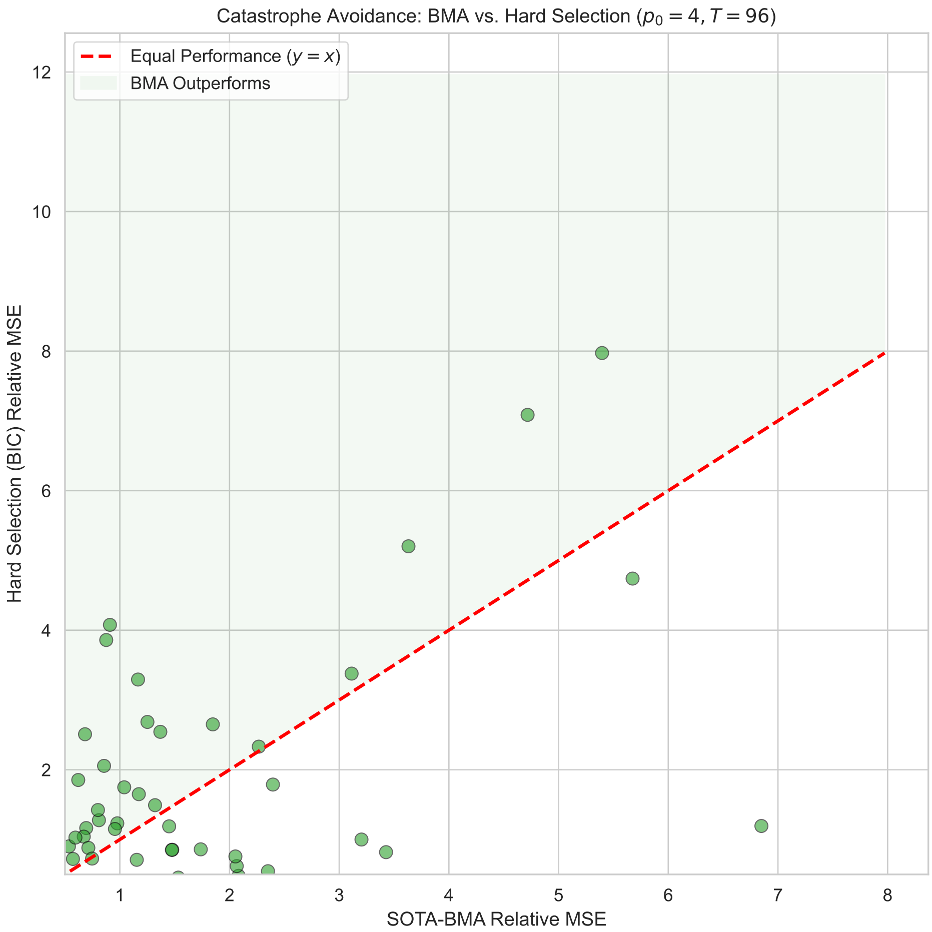

Figure 4: Catastrophe Avoidance: BMA vs. Hard Selection (BIC). Points above the 45-degree line indicate iterations where BMA outperformed BIC, notably smoothing out catastrophic tail risks.

Table 3: Relative Mean Squared Error: Hard Selection vs. Bayesian Model Averaging.

T	p_0	AIC	AICc	SIC (BIC)	HQC	OLS BMA (BIC-W)	Hybrid-BVAR (WN) (Fixed τ)	Hybrid-BMA (WN) (Fixed τ)	SOTA-BMA (WN) (Opt τ)
96	4	1.328	0.957	0.985	0.998	1.080	1.475	1.206	0.933
	6	1.426	0.913	0.886	1.028	**0.744**	1.574	1.456	1.001
	8	1.267	0.768	0.807	0.875	**0.673**	1.678	1.256	0.857
	10	1.283	0.855	0.829	0.800	0.839	1.272	1.276	**0.782**
144	4	1.106	1.206	1.145	**1.085**	1.165	1.962	2.043	1.245
	6	1.014	1.017	1.086	1.050	1.004	1.261	1.558	**0.984**
	8	0.908	0.918	0.920	**0.869**	0.985	1.168	1.018	1.011
	10	0.805	0.826	0.730	**0.719**	0.847	1.004	1.151	0.847
240	4	1.052	1.042	1.067	1.056	1.050	1.134	1.494	1.020
	6	1.211	1.018	1.026	**1.000**	1.253	1.475	1.265	1.321
	8	0.986	1.030	0.943	0.968	0.936	0.953	1.115	**0.915**
	10	**1.062**	1.236	1.289	1.249	1.761	1.139	2.212	1.312
480	4	0.956	1.589	1.214	1.091	0.978	1.428	**0.782**	1.248
	6	**0.905**	1.148	1.210	1.107	1.061	1.144	1.059	1.262
	8	0.999	1.218	1.307	1.122	0.985	1.328	1.069	0.980
	10	0.939	1.008	1.157	1.036	**0.930**	1.205	1.567	1.210

Table 4: Posterior Lag Weight Distribution for BMA ($p_0 = 4$).

Estimator	Hybrid-BMA (OLS W)	SOTA-BMA (MDD W)	Hybrid-BMA (OLS W)	SOTA-BMA (MDD W)	Hybrid-BMA (OLS W)	SOTA-B (MDD
T	96	96	144	144	240	240
Statistic / Lag						
$E[p]$	4.18	1.00	4.43	1.00	5.02	1.04
p=1	19.5%	100.0%	16.8%	100.0%	13.1%	95.8%
p=2	18.5%	-	16.9%	-	14.2%	4.2%
p=3	14.4%	-	14.1%	-	12.7%	-
p=4	11.2%	-	11.6%	-	11.2%	-
p=5	8.7%	-	9.3%	-	9.6%	-
p=6	6.7%	-	7.5%	-	8.3%	-
p=7	5.3%	-	6.1%	-	7.2%	-
p=8	4.2%	-	5.0%	-	6.2%	-
p=9	3.5%	-	4.1%	-	5.3%	-
p=10	3.0%	-	3.4%	-	4.6%	-
p=11	2.6%	-	2.8%	-	4.0%	-
p=12	2.4%	-	2.3%	-	3.5%	-

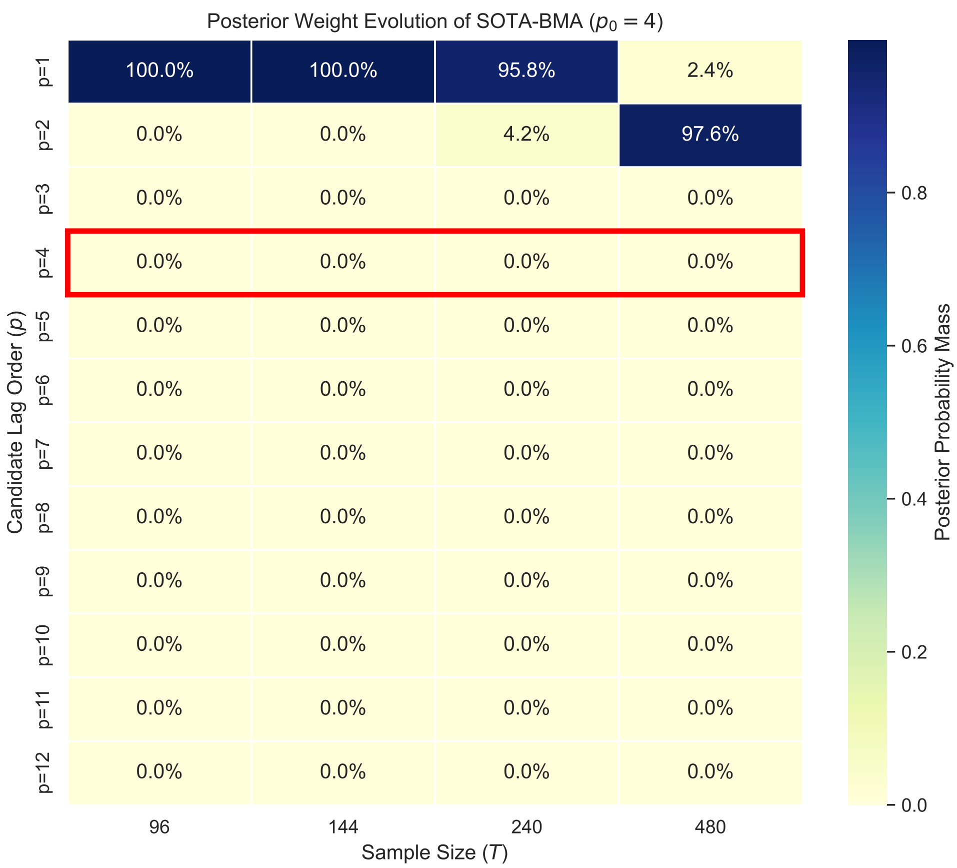

Figure 5: Posterior Weight Evolution of SOTA-BMA. At small sample sizes, probability mass is dispersed to hedge uncertainty. As T grows, the distribution concentrates precisely on the true DGP.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.